1. FluviSAR: Web Observatory

Tier 1: Core features

These define whether FluviSAR convincingly fulfils the challenge.

W1. Earth Change Atlas

An interactive global map with selectable investigation sites, including the Jamuna River as the flagship case. Each site opens a focused change investigation, with its area of interest, detected change zones, geographic context and available observation dates.

The atlas should demonstrate that the concept can travel beyond one location, while the primary story remains focused.

W2. SAR Time Machine

A date-driven comparison interface with synchronized before-and-after maps, a swipe divider, timeline playback and selectable observation pairs. The visualization progresses through the observation sequence rather than simply toggling between two screenshots.

Display acquisition dates and elapsed time between observations. This is the primary visual demonstration of change.

W3. Radar Lens

Let users move between the available SAR backscatter layers, polarization channels and derived change products. Include a proper legend, measurement units, selected acquisition, sensor information and a short explanation of what the signal represents.

For the flagship river case, users should be able to examine changes in radar response alongside the derived land, water or exposed-sediment classification.

W4. Change Detection Engine

Visualize the location and spatial extent of change candidates, not merely an image difference. For river erosion, include historical and current river boundaries, candidate retreat zones, accretion zones, water extent and exposed sediment.

Clicking a zone should reveal area changed, affected dates, change category and its position relative to the previous river boundary.

W5. Evidence and Confidence Inspector

Every selected change zone opens an evidence panel explaining the observation dates, source product, processing method, supporting layers, quality flags and uncertainty.

Show why the system considers the zone a change candidate, what evidence supports that interpretation and which alternative explanations remain plausible. Avoid presenting one unexplained confidence percentage as scientific proof.

Tier 2: Differentiating features

This tier is where FluviSAR moves beyond a conventional remote-sensing dashboard.

W6. Break My Finding: Change Robustness Laboratory

The signature feature. Give users control over selected thresholds, comparison dates and classification settings. Recalculate the displayed change map and show which regions remain stable, disappear or become ambiguous under alternative settings.

A selected erosion hotspot should have a visible robustness profile, not just a colored polygon. Users should be able to test whether the conclusion survives reasonable analytical changes.

W7. River Memory

Display multiple historical river boundaries simultaneously. Animate channel migration, changes in water extent, exposed sandbars and the progression of erosion and accretion.

Selecting a bank segment reveals its history and a plot of measured or simulated retreat through successive observations. This is more informative than a generic timeline because it gives a particular location a temporal history.

W8. Environmental Context Fusion

Let users compare SAR-derived change with contextual layers such as precipitation, river level, elevation, vegetation, land cover and optical imagery where suitable data are available.

Show whether each layer supports, complicates or cannot resolve the change interpretation. Temporal alignment matters: context from an unrelated date should not appear to validate a finding.

W9. Exposure and Impact Lens

Overlay settlement locations, cropland, roads, schools and other relevant assets. Show how much of each asset class intersects the detected change zone or a defined proximity buffer.

Present these as exposure estimates, distinct from verified damage or predictions of future losses.

W10. Change Passport

Generate a shareable evidence record for each investigation: source products, acquisition dates, map layers, processing settings, measurements, sensitivity results, contextual evidence, unresolved questions and field observations.

It should be the permanent bridge between FluviSAR's analysis and TRACE Field's investigation.

Tier 3: High-impact experience

W11. Radar-to-Reality

A linked, synchronized view of the original radar layer, classified change, contextual imagery and field evidence. Clicking one location highlights the same location in every panel.

The user sees how the system moves from radar observation to a mapped interpretation.

W12. Scientific Story Mode

A guided, cinematic investigation: select a river segment, observe the radar sequence, reveal the change map, inspect the evidence and show the possible human consequences.

The story is generated from the selected investigation rather than being a prerecorded animation unrelated to the interactive map.

W13. Scenario Laboratory

Explore hypothetical changes, such as alternative river positions or different water extents, and compare their spatial consequences. Clearly separate hypothetical scenarios from observed or derived change maps.

This adds an interactive exploration layer without confusing a scenario with a forecast.

W14. Shareable Investigation

Export a compact visual report or shareable investigation link with the location, dates, key finding, evidence layers and Change Passport. Include a QR code to open the corresponding case in TRACE Field.

2. TRACE Field: Mobile Companion

TRACE Field should not be a smaller copy of FluviSAR. Its purpose is to connect a satellite-derived finding to a structured field observation.

Tier 1: Core features

A1. Field Mission

Open a case from a FluviSAR share link or QR code. Display the target zone, investigation question, relevant dates, map and the expected field observations.

A2. Evidence Capture

Capture photographs, GPS coordinates, timestamps, notes and structured observations. Each observation is classified as supporting, contradicting, inconclusive or not observable.

The field record should remain linked to the particular change zone that prompted the investigation.

A3. Guided Field Verification

Provide a structured checklist for riverbank position, waterline, exposed sediment, vegetation, damaged embankments or nearby assets, depending on the investigation.

The objective is to collect evidence that directly addresses the satellite interpretation, rather than just upload generic field photographs.

A4. Offline Mission Pack

Save the case, essential map information and investigation form locally. Queue field observations for synchronization once connectivity returns.

This is particularly relevant to remote riverbank locations where consistent connectivity cannot be assumed.

A5. Evidence Record and Sync Status

Track each observation from capture to synchronization and review. Preserve its coordinates, time, observer attribution, media and relationship to the original satellite-derived finding.

Tier 2: Strong differentiators

A6. Navigate to the Change Zone

Show the user where the selected anomaly or erosion candidate lies, the distance to it and the target boundary. Provide a simple map-based way to record observations against the target location.

A7. Observation Quality Context

Display GPS accuracy, distance from the target, observation time, missing evidence and the relation between the photograph and the mapped zone.

This is a quality assessment of the submitted observation, not a probability that the satellite finding is true.

A8. Evidence Return Loop

Once synchronized, field observations appear in the corresponding FluviSAR case. Reviewers can inspect the evidence and update the status of the original finding.

This is one of the most important features across the entire two-product system.

A9. Multi-Observer Evidence

Let multiple observers submit evidence to the same case. Display conflicting observations separately rather than silently merging them into one conclusion.

Tier 3: Immersive field experience

A10. Field Mission Replay

Reconstruct an investigation chronologically: case opened, target reached, observations recorded, photographs captured, notes added and evidence synchronized.

A11. SAR Field Guide

A context-aware guide explains what a radar-derived change candidate means, what the observer should look for and which observations could support or contradict the interpretation.

A12. Field-to-Story Export

Compile selected satellite imagery, change polygons, field photographs, coordinates and observer notes into a compact visual report that can be shared from the mobile app.

Web

1. River Memory: the living change canvas

What it does

A full-screen geospatial experience in which the user moves through actual NISAR acquisition dates. The radar-derived river surface changes in synchrony with measured bank or water boundaries, revealing land-to-water transitions, water-to-land transitions and areas that remain ambiguous.

The map should retain faint outlines of previous observed boundaries. Selecting a patch opens its measured area, dates, source product and change classification.

Why this earns its place: Temporal visualization is explicitly encouraged by the challenge, and the Jamuna's complex river morphology gives it a compelling visual story. But simple before/after sliders already exist; the feature only becomes distinctive when the temporal views are tied to reproducible measurements and a clear interpretation.

Success criterion: Every date, boundary and number displayed must correspond to the underlying data. Do not interpolate observations into dates when no acquisition exists.

Web

2. Break My Finding: the counterfactual stress test8

What it does

The user selects a suspected change patch and deliberately challenges the initial classification.

The interface exposes four tests:

- Threshold sensitivity: recalculate the change mask across documented, reasonable classification thresholds.
- Hydrological confounding: inspect available river-stage observations and determine whether the acquisition pair has a material stage difference.
- Data quality: expose invalid pixels, coverage gaps, alignment issues and relevant product-quality flags.
- Independent comparison: compare the result with suitable external observations, when temporally and spatially compatible data are available.

The result is a spatially explicit breakdown of which portions remain stable across the tested settings, which depend strongly on the selected threshold, and which cannot be evaluated.

This is the most important differentiator I would add. It converts a passive visualization into an investigation. A judge can actively challenge a finding and see how the evidence responds.

The novelty is the integrated, user-facing workflow—not the isolated use of sensitivity analysis, which is already an established scientific method.

Success criterion: Changing a parameter must change the calculated mask and associated measurements. Do not present a canned sequence or label robustness as a probability of correctness.

3. The Change Passport: every claim has a traceable origin

What it does

Every selected finding has an evidence record that connects the displayed result to the actual processing inputs.

It includes the NISAR product and granule identifiers, acquisition dates, product maturity, available polarization, processing parameters, spatial reference, calculated area, relevant quality checks, supplementary evidence and unresolved limitations.

The record can be exported as a report plus a machine-readable manifest of the inputs and processing configuration.

Your existing FluviSAR processing documentation already describes a pipeline and data provenance. Retain those components if they generate the displayed results, and connect the output directly to the visual investigation rather than keeping provenance hidden in a separate methods section.

Why it matters: A judge should be able to select one striking patch and answer, “Where did this number come from?” without having to trust the UI or an AI-generated explanation.

TRACE Field

4. Field Proof: observations that can challenge the satellite

What it does

The user opens a case through a QR code or shared case link. TRACE shows the target finding, relevant acquisition dates and a simple observation workflow.

The observer records whether the visible scene supports the finding, contradicts it, cannot be observed or remains uncertain. The submission can contain a photograph, timestamp, GPS coordinates, reported GPS accuracy and a note.

The app must work offline for prepared cases. Once connectivity returns, it synchronizes the submission to the relevant case in FluviSAR.

FluviSAR shows the new observation as evidence awaiting review; it does not automatically convert a submitted photograph into a verified scientific finding.

Why it matters: The mobile app now has a distinct purpose. It closes the loop between satellite interpretation and observations made by people, instead of replicating the website's analytical controls.

Success criterion: Demonstrate a real offline submission and successful synchronization. If your team cannot obtain observations from the study area, label any demonstration record as sample or simulated input rather than claiming independent field validation.

P2 — add after the core works

TRACE Field

5. Feel the River: data-driven sonification and haptics

Turn actual measurements into an optional sensory experience on the phone.

For example, use a defined mapping from measured backscatter change or classified surface-transition magnitude to an audible parameter; use haptic intensity for a separate, documented measurement. Keep the numeric measurement visible beside the sound or vibration.

This can make a selected river reach feel like a changing instrument, rather than presenting another layer of colored pixels. It also connects directly to the challenge's metaphor of Earth's surface in motion.

The constraint: Sonification itself is not unprecedented, and the sound must not masquerade as a physical sound recorded from space. It is a deliberate encoding of measured data.

THE CHALLENGE'S DNA — Verbatim Hooks → Winning Features

1. *"the endless dance of natural processes and human activities… difficult to visualize, understand, and communicate"*

This is the challenge's thesis. The word **communicate** is the tell: they don't want an analysis tool, they want a *translation* tool. Radar is alien to normal humans — your job is to make the alien legible.

Winning feature: The Human Translation Layer — every technical output must have a plain-language twin. Not a tooltip, a twin. "ΔdB −2.4" sits next to "3 football fields of bank gone this pass." Your Bangla voice report and sensory/haptic modes already aim here. This clause is why they exist. Every screen: one number, one sentence a villager understands. Judges score this clause every time they look at a screen.

2. *"track and visualize one or more types of surface change at locations around the world"*

Two judging tests buried here: (a) *track* = time series is mandatory, not decorative; (b) *locations around the world* = plural, global ambition. A single Bangladesh case risks reading as "class project."

Winning features:

- **Time-Warp Dial** — the tracking instrument. Scrub real acquisition cadence (~6-day effective revisit), polygons grow, land counter ticks. This is the "track" receipt.
- **World Tour mode** — one global switchboard: your deep Jamuna case + 2–3 satellite cases elsewhere (wildfire burn scar, glacier terminus, subsiding city). Same engine, different physics per case — this is where Part 3's mode-specific signatures pay off: burn scar shows cross-pol collapse, glacier shows coherence loss at the calving front, subsidence shows fringes at 11.9 cm/2π. Three change types = you visibly exceed "one or more."

3. *"radar data is most sensitive to two things: roughness and the dielectric constant"*

They handed you the physics exam key. A submission that never demonstrates roughness or dielectric response missed the assignment.

Winning feature: The Physics Twin panel — for any detected change, show *why the radar saw it*: "σ° dropped 4.1 dB → surface smoothed (roughness ↓) and water rose (dielectric ↑)." Your polarization explainer (HH/HV/VH/VV) becomes the star here: wet soil = high dielectric

= dark; dry soil = brighter; flooded = darkest. One sentence per driver, on the map, at the pixel. This is the clause that separates science teams from UI teams.

4. *"hypothesize the physical drivers behind what you see (or don't see)"*

Note: **"(or don't see)"** — they explicitly reward explaining *absence* of signal. Almost nobody does this.

Winning feature: The Null Result Room — "NISAR saw NO change here. Why?" Three cards: (a) change happened but radar geometry missed it (look direction, shadow), (b) change too small for the resolution cell, (c) no change — stability confirmed. This feature is nearly free to build and nearly impossible for competitors to think of. It also feeds your Devil's Advocate Engine.

5. *"Which polarizations highlight these changes on land and ice?"*

Polarization is asked about *by name*. A winning project must have an interactive polarization answer.

Winning feature: Polarization Switchboard — the map re-renders per polarization with one tap, and each band answers its own question: VV floods, VH deforestation/roughness, HH structure. Caption on the switchboard: "Same river. Four truths." Fold in your HH/HV/VH/VV explainer. On the ice case, show HV revealing crevasse structure VV hides.

6. *"download supplemental data such as optical images or GIS layers to contextualize"*

"Supplemental" = expected, scored as thoroughness.

Winning feature: your **Cloud Autopsy swipe** (optical vs SAR) is the perfect response — but frame it as *contextualization*: the optical image is the context, the SAR is the truth. Add one ancillary GIS layer (population or settlement footprints) feeding the Impact Ledger.

7. *"intuitive tools and innovative visualizations... make NISAR data accessible to a broad audience"*

Broad audience is plural on purpose: scientist AND student AND decision-maker AND public.

Winning feature: Audience switcher — one toggle: ANALYST / DECISION-MAKER / PUBLIC. Same case, three vocabularies (dB & coherence / hectares at risk & confidence / "your char lost land, danger high" in Bangla). This single feature visibly answers "broad audience" four times over and costs almost nothing — it's the same manifest rendered three ways.

8. *"show how human drivers… are impacting Earth's natural resources"* and *"personal context and stories"*

The challenge invites the hometown story explicitly.

Winning feature: your **Personal Testimony layer** — a photo/voice note pinned on the map from the field ("my grandfather farmed this char") tied to the SAR verdict. One human moment, georeferenced, next to the algorithm's output. This is the emotional payload of the video and the answer to the human-drivers clause (sand mining, embankment engineering as erosion drivers).

9. *"What future applications of NISAR are possible?"*

Winning feature: a single **Future Missions horizon strip** in the app: dam deformation watch, landslide early warning, mangrove carbon monitoring — each a one-line card tagged "NISAR-ready." Cheap, and it proves you understood the instrument, not just the dataset.

10. *"present your findings in… a final report, video, presentation slides, story map — even an interpretive dance?"*

They signal format flexibility — so your **4-minute video + live app + short whitepaper** trio covers three formats at once. The cheeky "interpretive dance" aside is a gift: your haptic Radar Walk is literally *feeling the radar dance*. Say that sentence in the video ("we let a villager *feel* the satellite's rhythm") and you've used their own metaphor against them.

11. *"open-source, user-friendly… actionable insights"*

Winning feature: the **Provenance Chain + one-command reproducibility** (already approved). "Actionable" = the output ends in a decision: your auto-generated Bangla/English briefing sent to the field inbox is the action. Video must show the briefing arriving on the phone, not just existing on the web.

Web (6): Cloud Autopsy swipe · Devil's Advocate Engine · Time-Warp Dial (acquisition-date-accurate) · Prediction Front (labeled scenario) · Provenance Chain · Impact Ledger with MEASURED/ESTIMATED/SIMULATED badges.

App (5): Offline case inbox · **Point & Prove** AR overlay · Haptic Radar Walk · Bangla voice reports · One-tap verdict.

WEB (the brain):

1. Cloud Autopsy swipe (optical context vs SAR truth)
2. Time-Warp Dial time-series with live land counter
3. Physics Twin panel (roughness + dielectric + polarization answers)
4. Devil's Advocate Engine + Null Result Room
5. Polarization Switchboard
6. World Tour mode (erosion / burn scar / glacier — three change types)
7. Audience switcher (ANALYST / DECISION-MAKER / PUBLIC)
8. Impact Ledger + auto-generated briefing
9. Provenance & reproducibility

APP (the hands):

1. Offline case inbox
2. Point & Prove camera
3. Haptic Radar Walk (the "dance" made literal)
4. Bangla voice reports
5. One-tap ground verdict

🖥️ **WEB PLATFORM — Final Feature Set (6 features only)**

#	Feature	Why judges gasp
1	**Cloud Autopsy Theater** — fullscreen Optical vs. real NISAR L-band swipe over the Jamuna. Left: Sentinel-2, 92% cloud cover, useless. Right: NISAR GCOV, same date, river channels razor-sharp. Auto-animating slider. One sentence caption: *"The monsoon hid*	Instantly communicates the mission. Uses REAL granules.

	this for 4 months. The radar didn't blink."	
2	**The Devil's Advocate Engine** — every detection runs 7 adversarial checks live on screen, and the UI *visibly argues with itself*: "Backscatter drop detected → CHALLENGE: could be moisture, not erosion → GPM says 42mm rain → downgraded to PROBABLE → FIELD TEAM DISPATCHED → CONFIRMED." Each check is a card flipping from red to green.	This IS your moat. Nobody else self-cross-examines.
3	**Time-Warp Dial** — one radial slider scrubs 12+ real NISAR passes (asc+desc = ~6-day effective revisit) and the erosion polygons grow year-over-year. Land loss counter ticks in hectares + "homes equivalent"	Directly shows NISAR's killer feature: temporal density
4	**Prediction Front** — ML forecast layer: "high-risk bank segments for next season," with honest confidence band. Tie to FFWC/BWDB river stage + GPM rain. Labeled clearly as scenario, not forecast (your	Shows you went beyond detection into *actionable insight* — rubric gold

	honest-language discipline — keep it, judges respect it)	
5	**Provenance Chain** — click any polygon → full lineage: granule IDs, ASF DOI, processing params, audit results, field photos. One "Reproduce This" button that downloads the notebook + data manifest	Scientific credibility. Open-source rubric.
6	**Impact Ledger (Sendai-aligned)** — homes at risk, people, cropland, $ exposure per reach — but EVERY number carries a source badge: MEASURED / ESTIMATED / SIMULATED. Your honesty-as-design becomes the brand.	Differentiates from every team that inflates numbers

Kill from web: SAR Lab's 14 chapters (move 2 simulators into "Learn" as Easter eggs), Intelligence page's ML circus (confusion matrix, ROC, SHAP, attention maps — judges don't believe unverifiable ML), Mission Briefing cinematics, chat widget, inspect-everything mode. Deep cuts. Trust me.

📱 MOBILE APP (PWA → APK) — Field Sentinel

The phone is not a shrunken dashboard. It's a tool for a person standing on an eroding bank with wet hands and 2G signal.

#	Feature	Why it's astonishing
1	**Offline-First Case Inbox** — sentinels get assigned cases,	Real field credibility

	everything cached locally; works in airplane mode; auto-syncs later (you have this — harden it, demo it on video with actual airplane mode)	
2	**"Point & Prove" Camera** — open camera, frame the bank; app overlays the satellite's erosion polygon as an AR-style outline on the live view + GPS distance-to-target. Snap photo → auto-matched to polygon → verdict updates	**This is the 'how did they think of that' moment.** Ground truth becomes a game of lining up the ghost outline with reality. It closes the orbital→ground loop *visually*.
3	**Haptic Radar Walk** (keep FEEL, repurpose) — sentinel walks the bank; phone vibrates intensity mapped to real dB change underfoot. Finding the erosion front becomes *felt*, not read. 60 seconds in the video, judges remember it forever	Genuinely novel sensory science communication

4	**Bangla Voice Reports** (keep, make offline TTS) — a villager presses one button: "Your char lost 3 bigha this month. Danger high." in Bangla	Real impact on real users. Rubric hammer.
5	**One-Tap Verdict** — giant YES/NO: "Is the water where the satellite said?" → feeds back into the Devil's Advocate Engine on web	The loop completes

Web application: the analytical surface

The web app is for scientists, reviewers, and judges. It should feel like a working instrument, not a dashboard.

Module 1: The Stage-Matched Pair Selector

What it does: The user selects an AOI (e.g., Chilmari on the Jamuna). The app queries a river gauge time series (FFWC or BWDB data, simulated if necessary) and a SAR acquisition archive. It finds the two dates with the closest river stage values, then displays them side by side.

Why this wins: This is the only feature in your entire current stack that a remote sensing scientist would call a methodological contribution. Most change detection compares two arbitrary dates and calls any backscatter drop "water." You are isolating permanent change from seasonal inundation. That is Creative at 20% and Feasible at 20%.

What the judge sees: A split screen. Left: SAR image from Aug 26, stage 19.42 m. Right: SAR image from Sep 19, stage 19.44 m. Below: a delta-stage badge that says "0.02 m

difference. Valid pair." A second pair with a 1.68 m difference is shown in red and labeled "Rejected: seasonal inundation, not permanent migration."

Module 2: Live Otsu Threshold with Histogram

What it does: For any selected SAR raster, the app computes the Otsu threshold live and displays the backscatter histogram. The user can drag the threshold line and watch the water/land classification change in real time.

Why this wins: Otsu is ten lines of Python. It is a real algorithm. It is defensible. It makes the classification transparent instead of a black box. When a judge asks "how did you decide what is water?" you show the histogram, not a hard-coded number.

What the judge sees: A bimodal histogram with a vertical line. Drag the line left: water area grows. Drag right: water area shrinks. A pixel count updates live. This is the "aha, I understand the method" moment.

Module 3: Change Mask with Uncertainty Band

What it does: After thresholding both dates in the matched pair, the app computes land→water (erosion) and water→land (accretion) polygons. Every polygon carries an uncertainty value derived from threshold sensitivity: run Otsu at ±2 dB and count how much the polygon area changes.

Why this wins: No naked numbers. Every hectare is reported as "1,247 ha ± 180 ha (95% CI)." This is the cheapest way to look more scientific than every other team. It signals that you understand measurement error.

Module 4: The Rejection Engine

What it does: A dedicated mode that walks through every reason a detection might be false: stage mismatch, radio frequency interference (RFI), wind roughening the water surface, emergent vegetation, swath edge artifacts, or missing cross-sensor confirmation. Each reason is a card with a pass/fail/warning status and a plain-language explanation.

Why this wins: This is your most distinctive feature. Most apps tell you what they found. Yours tells you what it rejected and why. The TRACE screenshots already show a "12% confidence, likely false positive" screen buried in a modal. Move it to the front. Frame the entire app as: *"A system that knows when it doesn't know."*

What the judge sees: A detection with a green "SUPPORTED" badge next to a rejection with a red "REJECTED: stage mismatch 1.68 m" badge. The rejection is not hidden. It is a first-class feature.

Module 5: Cross-Sensor Evidence Wall

What it does: For any detection, the app shows independent evidence from other sensors: Sentinel-1 water extent (OPERA DSWx-S1), GPM IMERG precipitation anomaly, NASA FIRMS thermal anomalies, and the river gauge record. Each source gets a card with status (agree/disagree/unavailable) and a confidence contribution.

Why this wins: The challenge explicitly invites using supplemental data. This makes the science real even when the primary NISAR granule is simulated. It also creates a failure mode you can show on stage: "OPERA unavailable for this date, confidence drops from 87% to 61%." A system that degrades gracefully is more credible than one that always says 94%.

Module 6: What Changed for Whom

What it does: For each erosion polygon, overlay WorldPop population data and OpenStreetMap infrastructure (schools, clinics, roads). Generate one sentence per polygon: *"This 42 ha erosion polygon contains 1 school, 0 clinics, and the main road to Chilmari."*

Why this wins: This is Impact at 30%. Not "45,000 people at risk" in a dashboard tile, but "the road to Chilmari is gone" in a sentence a district officer can act on.

Mobile app (PWA): the field surface

The mobile app is for the person standing on the riverbank. It should feel like a field tool, not a shrunken dashboard.

Module 7: Offline-First Field Verification

What it does: The field worker opens the app in airplane mode. The app shows the pre-loaded detection polygons. The worker taps a polygon, sees the satellite's claim ("this was land, now water"), and responds with one tap: Confirmed or False Positive. They can attach a photo and a note. When connectivity returns, the observation syncs and the detection's confidence updates.

Why this wins: This closes the loop between orbit and ground. The challenge asks for actionable insights. A field worker confirming a satellite detection with a photo is the most actionable thing in your stack. The TRACE report already describes this feature. Make it the centerpiece of the mobile demo.

Module 8: Sonification and Haptics (FEEL)

What it does: Map the backscatter delta to audio frequency and haptic rhythm. A strong erosion signal (−3 dB or more) produces a low, slow pulse. A weak signal produces a high, fast click. The phone vibrates in sync. The field worker can "hear" the change without looking at the screen.

Why this wins: This is your Creativity at 20%. No other team will have this. Justify it with one sentence: *"For field workers in low-light or hands-busy conditions, and for visually impaired researchers, the delta-dB is rendered as audio and haptics instead of a visual map."* Do not oversell it as a primary interface. Sell it as an accessibility and field-convenience feature.

Module 9: The Decision Brief Generator

What it does: One button generates a one-page PDF: finding, confidence, affected population, recommended actions, contact for local authorities, QR code to the field form. The PDF is the artifact you hand to a judge.

Why this wins: Judges remember what they can hold. A PDF that says *"1,247 ha ± 180, 8,200 people, 3 schools, one road, recommended action: relocate the Chilmari market"* is more persuasive than any dashboard. It is also the Sustainability criterion: the app produces something a real decision-maker can use after the hackathon ends.

Web (FluviSAR): the laboratory

Tier	Feature	What the judge sees	NASA ask it covers
1	**Time-series explorer**	Slider through every NISAR pass for the hero reach, with a per-date water mask and an area-vs-date curve	Different dates, time steps, interactive

Tier	Feature	What the judge sees	NASA ask it covers
1	**Polarization lens**	HH and HV side by side, with histograms for water, sand and vegetated bank	Which polarizations highlight change
1	**Hypotheses board**	Each driver (stage, rainfall anomaly, bank material, upstream flow) with evidence for, evidence against and a test	Hypothesize physical drivers
1	**Context layers**	Toggle Sentinel-2, population, land cover and settlements under the radar	Optical and GIS context, human drivers
1	**Provenance and open repo**	Click any number to see its granule, script and DOI, plus a one-command reproduce	Open-source, trustworthy
1	**Future NISAR panel**	What the tool does as the archive grows toward the full record	Future applications
2	**Equal-Stage Twin**	A stage ladder, one dot per pass at its river height. Drag two dots to equal height and the true change appears	Innovative visualization
2	**Consensus Core**	A map of pixels that stay “erosion” across many threshold settings, with the area shown as a range	Scientific validity
2	**Claim Ledger**	Every claim tagged Measured, Cross-checked, Unverified or Refuted	Trust, broad audience
3	**Sentinel-1 vs NISAR**	The same reach in C-band and L-band	Different frequencies
3	**Real InSAR case**	One earthquake case (GUNW), shown as fringes then centimeters	Earthquakes, global
3	**Cloud race**	Real Sentinel-2 cloud cover per date vs radar availability	Why radar
3	**SAR Lab**	Your educational simulators, titled “Educational simulation”	Educators

App (TRACE): the field instrument

Tier	Feature	What it does	NASA ask it covers
1	**Field verification**	GPS, photo, one-tap answer, note, offline queue and sync	Actionable, usable by the public
1	**Offline radar tile pack**	Real backscatter and change layers cached on the phone, with your position on them	Real NISAR data in the app
1	**Ground Biography**	Tap or stand anywhere to see that pixel's state across all passes ("river on Sep 7, land since Sep 19")	Time steps, personal context
2	**Satellite Asked Me**	The web's contested pixels arrive as field tasks, and an answer updates the web verdict with an audit trail	Connected platforms
2	**Walk on the Radar**	Your GPS reads the real pixel's backscatter and plays it as pitch and vibration	Innovative, accessible
2	**Verdict feedback screen**	Shows "your answer moved this case from X to Y"	Closing the loop
3	**Bangla voice and pictograms**	Low-literacy field use	Broad audience
3	**Next-pass countdown**	When NISAR will next image your location, from the 12-day cycle	Future applications

FluviSAR (web): the laboratory

1. **Equal-Stage Twin.** Replace the date slider with a stage ladder, one dot per NISAR pass at its river height. Drag two dots to the same height and the change map appears. It teaches why a flooded river is not eroded land.
2. **Consensus Core.** Run your pipeline across a grid of thresholds, filter windows and stage tolerances. Show per-pixel agreement: a solid "robust core" (erosion under every setting) and a faded "contested fringe." Report the area as a range. This replaces the fake confusion matrix with real uncertainty and turns your 8× swing into the headline.

3. **Sensor Race with real clouds.** Pull actual Sentinel-2 cloud cover for each NISAR date over your reach, for example via a STAC catalogue. Show "optical had a clear view on N of M days, radar on all of them." It replaces the simulated slider.
4. **One real InSAR case.** NASA's release shows NISAR-derived ground displacement for the June 24, 2026 Venezuela earthquakes, using passes before and after. Run one real GUNW case to cover the challenge's earthquake, subsidence and centimeter-scale movement themes, and delete the synthetic fringe toy. Check the product availability and ASF's known-issues page first.
5. **Claim Ledger and "run it on any reach."** Every claim is tagged Measured, Cross-checked, Unverified or Refuted, with a manifest hash. Publish the repo with a one-command rerun.

TRACE (app): the field instrument

1. **"Satellite Asked Me" tasks.** The contested fringe becomes map pins. The user goes there, taps one answer (water, land, sand or crop), and takes a photo. The web verdict then moves with the audit trail. This is your climax demo.
2. **Walk on the Radar.** Using GPS, the phone reads the real backscatter value of the pixel you stand on (from a pre-cached tile) and plays it as pitch and vibration. Walking toward the water, you hear the radar fall silent. It also works for low-vision users. This is the real-data upgrade of FEEL.
3. **Ground Biography.** Tap anywhere to see the pixel's history across all NISAR passes: "Water on Sep 7. Land since Sep 19. You're standing on ground that was river 24 days ago."
4. **Offline and Bangla voice.** Show these briefly as reliability proof.

WEB — TRACE Atlas (see it, dissect it, explain it). 7 features only.

W1. Rhythm Player — Not a timelapse. Pick a site, press play: the map animates through NISAR dates while a synced rhythm graph below plots σ^0 backscatter per pass, with river

stage / rainfall plotted on the same axis. You literally *see* the monsoon pulse, the glacier's seasonal beat, the burn scar's recovery curve. *Science: GCOV time series, 12-day NISAR repeat. Answers NASA #4 — patterns and rhythms.*

W2. Polarization Prism — One scene, one rotating prism control: HH → HV → HH/HV ratio → Pauli RGB composite. Each turn reveals a different physical structure — surface scattering, volume scattering, double-bounce — with the panel stating plainly: *"HV exposes the forest canopy change. HH exposes the flooded vegetation. On ice, HH/HV separates the calving front." Science: NISAR dual-pol GCOV covariance terms, L-band scattering physics. Answers NASA #6 — the polarization question, word for word.*

W3. Frequency Duel — Same place, same date, three panels racing: Optical vs Sentinel-1 C-band vs NISAR L-band (plus NISAR S-band where available). A cloud-density slider kills the optical panel first, weakens C-band under canopy, L-band keeps dancing. Penetration-depth animation under the forest canopy shows *why*. *Science: wavelength vs penetration, dielectric constant. Answers Potential Consideration #1 — different polarizations AND frequencies.*

W4. Driver X-Ray — "Why did it dance?" — Click any change polygon and get the causal chain, not just a colour: Driver (GPM rainfall +42mm / river stage / FIRMS fire / seismic event) → surface physics (dielectric up, roughness down) → radar signature (ΔdB −6.1, coherence lost) → verdict. Every change on the platform must pass through this panel. *Answers NASA #6 — hypothesize the physical drivers. This is your FluviSAR evidence trail, sharpened into one feature.*

W5. Fringe Stage — the deformation dance — The one dance amplitude maps can't show. Interactive InSAR interferogram for an earthquake/subsidence site: drag across fringes, each fringe counts 11.9 cm of line-of-sight movement (half the L-band wavelength), total displacement reads live; coherence map toggles beside it. *Science: GUNW/GSLC phase. Covers the challenge's earthquake + subsidence change type — 90% of teams will ignore phase entirely.*

W6. Five Dances Story Stage — The homepage is not a dashboard, it's a story map: Jamuna flood/erosion (your hometown chapter, personal context), Glacier Bay ice retreat, NorCal wildfire + recovery, Sundarbans wetland loss, one agriculture chapter (crop + soil-moisture swelling/shrinking, SME2). Each chapter carries its human-driver layer — population, farmland, settlements on the map — so the dance always lands on people. *Answers NASA #5 + #7 — ancillary data, human drivers, storytelling.*

W7. Future Dance Simulator — Close the loop NASA explicitly opens: *"What future applications of NISAR are possible?"* Set a future condition — +1m stage, drought year,

second burn — and the simulator renders what NISAR *will see*, generated from the Driver X-Ray relationships (stage → water extent → ΔdB), clearly staged as simulation. Ends with the future-applications board: disaster response through cloud, carbon/biomass, deformation early warning. *Answers NASA #6, second half.*

APP (APK) — TRACE Field (feel it, verify it, watch your place). 6 features only. No analysis — a completely different job.

A1. FEEL the Dance — Drag a finger across the SAR scene: water hums low, vegetated char buzzes mid, eroded/changed ground warns high — audio + haptics mapped deterministically from ΔdB. A blind user, a student, a farmer in bright sunlight can *read* the radar by touch and sound. This is your Art & Technology / accessibility shot — keep TRACE's FEEL, it's your most original asset.

A2. VERIFY — Ground Witness — A mission arrives from the Web ("this polygon changed — is it true?"). Go there: GPS locks, one-tap YES / NO / UNSURE, photo, offline queue, syncs when network returns. Your existing real backend feature — keep as is.

A3. My Place Watch — The hometown feature. Drop a pin on your village/char/farm once. The app builds *your* dance card: your place's time-series animation, its current flood/soil-moisture status, its driver explanation in plain Bangla. Personal beats global for the public audience.

A4. Pass Countdown + Dance Alert — NISAR revisits every 12 days. The app counts down to the next pass over My Place, then alerts: *"Your river danced — 3 new change polygons near your pin"* with the before/after in one swipe. The satellite's rhythm becomes a phone habit.

A5. Pocket SAR Lab — For the educator audience NASA names: 4 thumb-sized experiments — Cloud Blindness Test, Polarization Match (match the pol to the change it reveals), Speckle Filter swipe, Fringe Counter. Each 60 seconds, each teaches one real SAR concept from the Web's science. Students arrive at the Web already understanding it.

A6. Field Story — Standing on a char, record a 60-second voice + photo story ("this was my father's land"). It auto-pairs with that location's SAR scene for that date and is published into the Web's Jamuna story chapter (W6). Personal context, supplied by the people living the change — NASA asks for exactly this and almost nobody will have real voices.